Example 8.13. A corner-covered board is a board of $2^n \times 2^n$ squares in which the square at one corner is covered with a single square tile. A triominoe is a L-shaped tile formed with three adjacent squares (see Figure 8.10). Cover the corner-covered board with the L-shaped triominoes without overlap. Triominoes can be rotated as needed.

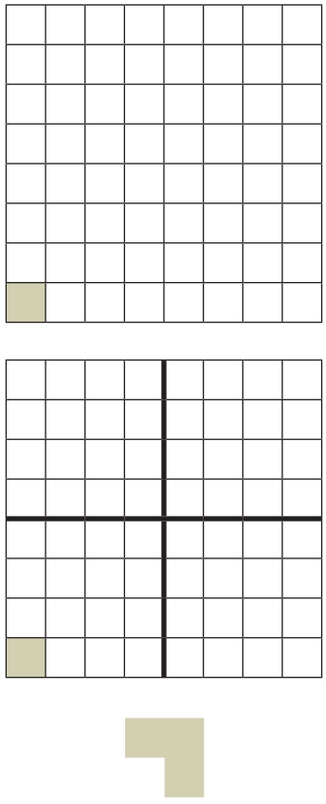


Figure 8.10: Corner-covered board and triominoe

The size of the problem is n (board of size $2^n \times 2^n$). We can solve the problem by recursion. The base case is n = 1. It is a 2×2 corner-covered board. We can cover it with one triominoe and solve the problem. In the recursion step, divide the corner-covered board of size $2^n \times 2^n$ into 4 sub-boards, each of size $2^{n-1} \times 2^{n-1}$, by drawing horizontal and vertical lines through the centre of the board. Place a triominoe at the center of the entire board so as to not cover the corner-covered sub-board, as shown in the left-most board of Figure 8.11. Now, we have four corner-covered boards, each of size $2^{n-1} \times 2^{n-1}$.

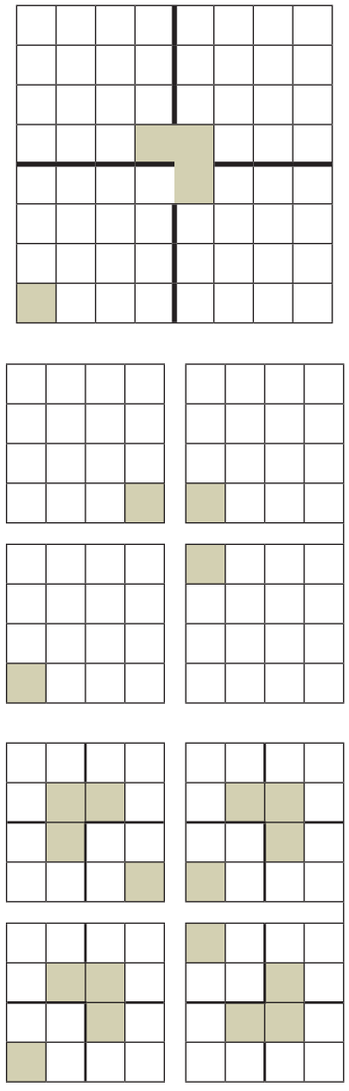
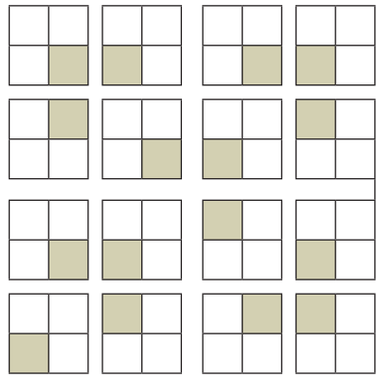


Figure 8.11: Recursive process of covering a corner-covered board of size 2×2^3



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